SMARTBRIDGE

SKILL WALLET
A SmartBridge Product

No. of Functional Features Included in the Solution

Date	28 June 2026
Team ID	XXXXXX
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S.No	Feature Name	Feature Description	Module / Componen t	Status	Marks Contribution
1	User Input Form	Allows users to enter Nitrogen (N), Phosphorous (P), Potassium (K), Temperature, Humidity, pH, and Rainfall values.	Frontend (HTML, Bootstrap)	Done	High
2	Input Validation	Validates user inputs before sending them for prediction.	Flask Backend	Done	High
3	Data Preprocessing	Cleans and prepares input data before passing it to the ML model	Data preprocessing Module	Done	High
4	Crop Prediction	Predicts the most suitable crop using the trained Random Forest model.	Machine Learning Module	Done	High
5	Prediction Result Display	Displays the recommended crop on the result page.	Frontend + Flask	Done	Medium
6	Model Persistence	Loads the trained machine learning model from the Pickle (.pkl) file.	ML Model Loader	Done	Medium
7	Error handling	Displays appropriate messages for invalid or incomplete input values.	Flask Backend	Done	Medium
8	Responsive User Interface	Provides a responsive interface compatible with desktop and mobile browsers.	HTML, CSS, Bootstrap	Done	Medium

Feature Summary:

Metric	Count / Value
Total Features Planned	8

Total Features Implemented	8
Core / Must-Have Features	6
Additional/ Nice-to-have Features	2
Features Tested & Verified	8

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	2	Input Form, Result Display
2	Backend / Logic	3	Input Validation, Data Preprocessing, Crop Prediction
3	Database / Storage	1	Pickle Model (.pkl), CSV Dataset
4	API / Integration	0	None
5	Security / Authentication	0	Not implemented in the current version